

Classroom learning dynamics using a cellular automata spatiotemporal model comparing peer instruction and traditional instruction

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(Dated: June 2025)

Peer instruction (PI) has become one of the popular means of classroom instruction in Physics Education. Such instruction method is vastly different from how classes are traditionally handled where the instructor conducts a lecture for the entire duration of the class. In this study, we model classroom learning as a probabilistic cellular automata model and investigate the effects of different factors such as seating arrangements, class size N , average learning rate λ , and heterogeneity $\delta\lambda$ on the overall class learning efficiency. The learning efficiency of the class is measured in terms of the class learning curve using the total learning time $t_{\max}$. PI mode appears to be more applicable with heterogeneous classes (high in-class learning rate difference $\delta\lambda$). As commonly known, TI mode is shown to be more advantageous for large N and homogenous class aptitude (low $\delta\lambda$). We show that the class learning curves of highly heterogeneous classes with TI mode have two stages of learning: a fast initial stage and a slow final stage. The slow final stage class learning rate is due to the late learners with lower λ -values. Such curves are absent with PI that instead consistently shifts uniformly. Our simplified model shows that PI mode performs better than, or at least similar to, traditional instruction. A mix of TI and PI would be the optimal mode of instruction taking advantage of TI for fast learning in the initial stage and the robust learning curve of PI in the later stages of class learning. Despite these simplifications, our model is able to provide general insights that are in agreement with previous studies and existing practices. We also introduce a class-level estimation of the Hake gain g based on a fitting recurrence function of the simulation data return map. Our approach gives consistent results with real-world observations for PI modes and provides opportunities for empirical determination of class-level gains.

CONTENTS

I. Introduction	1
II. Cellular Automata Method	2
III. Results and Discussion	3
III.1. Dynamical difference between TI and PI	3
III.2. Transmission and heterogeneity	4
III.3. Combined effects	5
IV. Comparison with available data	5
V. Summary and Conclusion	7
References	7

I. INTRODUCTION

Peer instruction (PI)—an increasingly popular method of instruction—attempts to solve a disconnect between students’ ability to answer quantitative and conceptual questions, particularly in physics subject matters [1, 2]. Such disconnect is said to be attributed to the typical approach of memorizing physics problem patterns and

applying the same solution patterns regardless of their potential non-applicability. The PI method forces students to think critically and to engage with peers guided by the learning material as opposed to simply absorbing the information from lectures given by the instructor as in the traditional mode of instruction (TI). The method’s effectivity has been shown not only in high school physics settings, but also in other disciplines and higher education settings [3, 4].

The PI method as outlined by Mazur and Somers [1] typically involves an instructor giving a short lecture on the topic, followed by a multiple-choice question (MCQ) that the student answer individually. PI emphasizes on the next phases of instruction which involve repeated cycle of students sharing their answers to the MCQ and discussing them with peers, usually until they get the concepts right. This part makes the PI method student centric requiring the instructor to intervene with slightly modified steps to adjust to the needs of the class. As long as the same learning objectives are tackled, the instructor can choose variations in the method [5]. The instructor may often decide on: increasing the multiplicity of the peer sharing and discussion phase; having different kinds of requirements and/or levels of evaluation [6]; providing different MCQ sets; or giving another short lecture to ensure clarity of concepts involved [7]. Modifications to the PI method have also been reported to minimize some undesired behavioral patterns such as just copying off each other [5]. The PI therefore contrasts significantly against the traditional method (TI) with which the in-

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structor simply gives a lecture for the entire duration, typically with minimal peer interactions among the students during class time.

Despite the unique approach of de-emphasizing actual problem-solving during the PI lecture phase, quantitative problem-solving skills have been consistently shown to remain uncompromised, and in many cases even improved, relative to TI. Several advantages were pointed out in favor of the PI method in relation to the following dimensions: decrease in the number of students with extremely low scores [6–8], improved conceptual understanding [6], as well as being independent of the background knowledge [7, 9].

The dynamics of the learning process involved during PI is complex resulting from the interaction during the peer discussion stage. Existing mathematical models for this are either predictive such as with the case of the artificial neural network (ANN) models [10, 11] or lack the spatial aspect of the process such as utilizing differential equations [12, 13]. Somewhat in between these is a generalized Ising model by Bordogna and Albano [14, 15] that considers three sources of information for the student to learn from: teacher instruction, peer interaction, and bibliographic materials (books, lecture notes, etc.). Their model shows that students learn more when they engage discussions with their peers than those who only listen to lectures. They also show that group structure affects student learning, and that low aptitude students may learn at the expense of high aptitude peers—a transmissionist view of PI. A survey of many other mathematical techniques are found in the work of Nitta [13]. We are yet to find mathematical models for learning that compares PI with TI.

Complementing these frameworks, actual classroom assessment results have also been used to model learning under PI. Roxas *et al.* [10] and Monterola *et al.* [11], for instance, trained ANN models to predict students' second assessment scores based on seating arrangements. Using this neural network, they were also able to characterize students' learning and investigate the effects of group homogeneity. The study revealed that an optimal seating arrangement exists for students under PI methods based on their final aptitude. The measure of each students' improvement was calculated via the Hake gain [16] as

$$g \equiv (S_2 - S_1)/(1 - S_1) \quad (1)$$

where S_1 and S_2 are scores for the first and second assessment, respectively. They also used the output/input ratio ($O/I = S_2/S_1$) to gauge student improvement. Class improvement is the average of any of the scores. The results of their study show that the outer corner seating arrangement (SA) performed the best using both the Hake gain and the O/I , followed by inner corner, then random, then center. The seating arrangements are illustrated in Fig. 1. In simulated classrooms (ANN system), each with 64 students and 10 classrooms in total, they found that homogenous classrooms with low aptitudes have significantly higher O/I -values under PI. This means that low aptitude students benefit the most from being grouped

together.

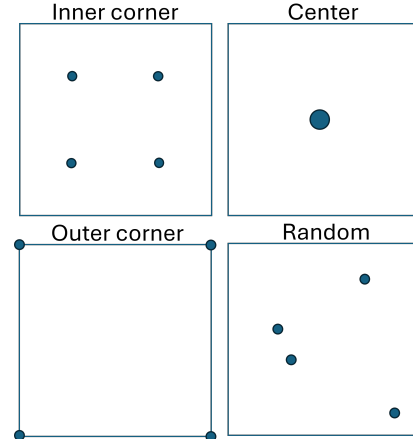


FIG. 1. Seating arrangements (SAs) considered for PI. Dots represent position of high aptitude students. The position of other students is typically arranged in rectangular pattern across the room (here square for brevity). The random case is just one of the many possible arrangements. Unless specified, we present the inner corner SA in our results.

Hake gain predicts the following expression for the expected score S_2 during the assessment after the first such that when done iteratively, every next assessment S_2 is dependent on the first assessment score S_1 as follows:

$$S_2 = S_1 + g \cdot (1 - S_1) \quad (2)$$

where g may be a function of S_1 . Nitta [13] presented a model of PI learning that relates the probabilities that the students answer the MCQ correctly (effectively, post- and pre-test). Such probabilities would then imply that the Nitta Model predicts that the gain is dependent on the previous score such that:

$$g = g(S_1) = S_1. \quad (3)$$

Other models provide Hake gain expressions (e.g. Pritchard *et al.* [12]) resulting to more complex formulation for the function $g(S_1)$. This model considers other factors that may affect the learning process such as knowledge before instruction and learning retention. Despite the diversity of the models that we found, current models of PI lack some aspects that we would like to incorporate together like seating arrangements, students' learning rate, and aptitude heterogeneity with which the introduction of PI from TI are typically founded on.

II. CELLULAR AUTOMATA METHOD

We use a probabilistic cellular automata (pCA) model [17] (also known as stochastic cellular automata [18–21]) to study the spatio-temporal dynamics of learning for a more direct implementation of any instructional mode. For this pCA model, each cell in the automaton represents a student arranged in such a way

each student is adjacent (a CA *neighbor*) to four others resulting to the usual rectangular lattice configuration. For the rectangular lattice, we assign the unique position (i, j) to each student such that $i, j \in [1, L]$ and the classroom size is $N = L^2$. The students at the edges and corners have less number of neighbors as in the real situations.

Each student at every lattice position (i, j) is then assigned a learning rate $\lambda_{i,j} \in \mathcal{R}$ as a quantification of how fast they learn as an individual with $0 \leq \lambda_{i,j} \leq 1$ (or a probability to learn from any other neighbor). This parameter is directly related to the learning aptitude of the students such that we consider the fast learners to have *high aptitude score* ($\lambda \gg 0$). We also introduce the state $s_{i,j}$ representing degree of knowledge of the student such that in this work we use $s_{i,j} \in \{0, 1\}$ as a two-level distinction between *learned* (or high-scoring student, $s_{i,j} = 1$) and *unlearned* (low-scoring, 0). The choice of two-level state should not prevent the possibility of having more levels as long as they can be quantitatively discriminated.

For each time step, $t \leftarrow (t + 1)$, each student's state $s_{i,j}$ is updated depending on the mode of instruction described in the following. We use an algorithm (see Fig. 2) general enough to contain various modes of instructions. The mode of instructions defines how $P_{i,j}$ is obtained.

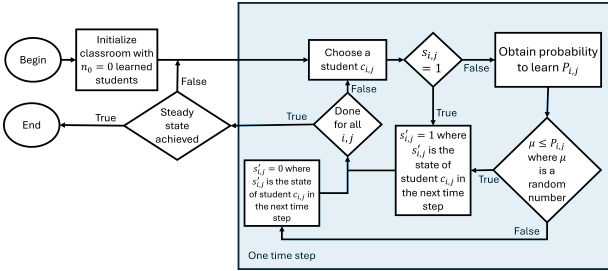


FIG. 2. Flowchart for the pCA algorithm. The PI and TI differ on how the probability $P_{i,j}$ is obtained as described in the text.

For traditional instruction (TI) mode, the probability of a student to learn in each time step ($P_{i,j}$) is

$$P_{i,j} = \lambda_{i,j} \rho_{\delta i, \delta j} \quad (4)$$

with $\rho_{\delta i, \delta j} \in [0, 1]$ as the potential dependence of learning efficiency on the relative position $(\delta i, \delta j)$ of the source of learning (inspiration, input, interaction, discussion, etc.) with respect to the student position (i, j) [22]. The relative position of the student may matter since instructors in TI mode, for example, may be dependent on the row i since usually students near the front ($i = 1$) tend to be more attentive and engaged in discussions than those in the last row ($i = L$). To further simplify for TI, we fix $\rho_{\delta i, \delta j} = \rho_0$ as the probability of the students to learn from the teacher directly as if during an efficient plenary (or broadcast) lecture throughout the class such that we simply have $P_{i,j} = \lambda_{i,j} \rho_0$.

Different from TI, the peer instruction mode (PI) involves some learning during interaction with peers. Such

interaction happens with the nearest neighbor unless specific instructions say differently. We consider the simple case such that the probability $P_{i,j}$ is dependent also on the states of the nearest neighbor. In this case, we have compounded probabilities resulting to

$$P_{i,j} = 1 - \prod_{\forall \delta i, \delta j} [1 - (\lambda_{i,j} \rho_{\delta i, \delta j}) (s_{i+\delta i, j+\delta j})] \quad (5)$$

where we now have dependence on the state $s_{i',j'}$ ($i' = i + \delta i$ and $j' = j + \delta j$) factored in every probability of learning transmission. We also consider the simplified case such that $\rho_{\delta i, \delta j} = 0$ for $|\delta i| > 1$ and $|\delta j| > 1$ and $\rho_{\delta i, \delta j} = \rho_0$ (fixed, isotropic). This assumption translates to the class keeping a seat arrangement in a lattice pattern and that peer instruction can be modelled as essentially nearest-neighbor interaction only (we use Moore neighborhood) [20, 23, 24], although higher degree neighborhood interactions are possible [22].

The model allows incorporation of class heterogeneity due to variation in the learning rates of the students such as by specifying the distribution function for random assignment of λ -values. A variation about a fixed rate $\lambda_0 = 0.5$ is chosen for this work such that $\lambda_{i,j} = \lambda_0 \pm \delta \lambda$ (two-level heterogeneity). The value of $\delta \lambda$ acts as the parameter that determines the degree of class heterogeneity as it is tuned from 0 to less than 0.5 (we are limited by the probability range $0 < \lambda_{i,j} < 1$). We have to remove the case $\delta \lambda = 0.5$ that makes the model unrealistic (having $\lambda = 0$).

III. RESULTS AND DISCUSSION

All TI simulations start with no learned students, while PI simulations start with four (4) learned students. Class performance is measured as the speed of learning. Differences in the dynamical development over time of the class overall state is also observed. Here, we use the time $t_{\max}$ (in arbitrary units, a.u.) it takes for the entire class to reach a steady state. This steady state is considered reached during the numerical experiments when no new student is recorded to have transitioned from *unlearned* ($s = 0$) to *learned* ($s = 1$). This consideration prevents from having unnecessary very long waiting time for the student of very low learning rate $\lambda \approx 0$ to transition $s : 0 \rightarrow 1$. The average quantities and standard errors are based on twenty(20) independent randomized runs.

III.1. Dynamical difference between TI and PI

Fig. 3 shows the typical monotonic increase in the fraction of learned students in classes starting with an unlearned state (all students in *unlearned* state, except for PI with which we start with four initially learned students). TI mode shows an early onset of high number of learning students in the first few iterations of lecture. The overall learning rate for TI tapers off as the entire class waits for the last few persons to actually learn.

Due to the required percolation between learned and unlearned states, the PI mode exhibits a gradual initial increase in the number of learned students, followed by a sudden sharp rise in the later stage. PI mode catches up fastest in the later part of the iterations especially for low class size N . This observation for the PI is consistent independent of the seating arrangement (SA) considered. Class size does not affect TI mode as expected from the broadcasted mode of learning where each student independently and simultaneously learns from the teacher directly.

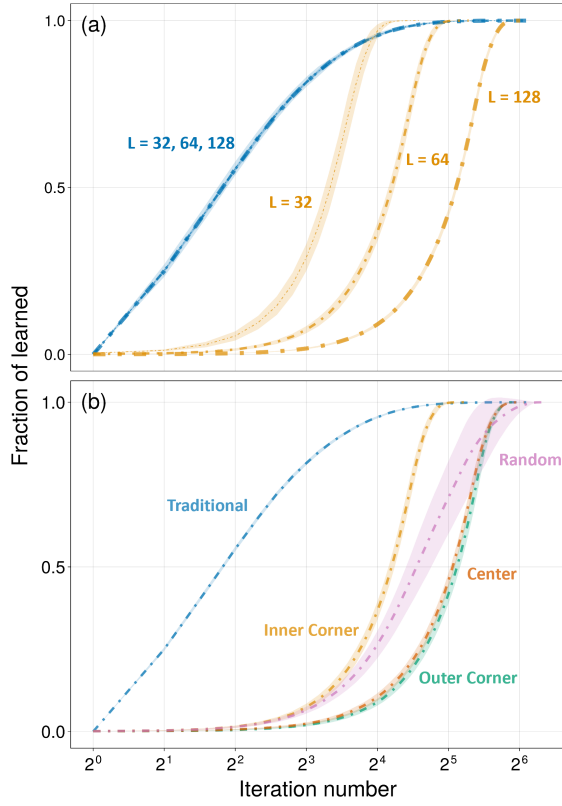


FIG. 3. Different monotonic increase of the total fraction of students in the learned state over iteration number. The blue colored plots are for TI while other colors are for PI. The same color corresponds to the seating arrangement (SA) used. Each PI's SA corresponds to a different color: orange (for inner corner), green (outer corner), orange (center), and pink (random). The line width is proportional to the N used. Other pertinent parameters fixed: $\rho_0 = 0.5$ and $\delta\lambda = 0.2$. The bands around each line show the standard deviation of the data after 20 independent random trials. Higher fraction of learned students indicate better overall learning. (a) Plot for different class sizes N . The best performant SA (Inner Corner) is used for comparison. (b) Plot for different seating arrangements (SAs).

Fig. 3(b) shows the effect of SAs for the PI mode. The inner corner SA outperforms all the other SAs for PI mode since such SA always has the highest rate of and earliest onset of rise in the number of learned students in the class compared to other SAs. When the students are randomly arranged, such SA performs as average of

the two extreme types. The center and the outer corners perform similarly due to the idealized assumptions of the model (e.g. isotropic co-learning).

III.2. Transmission and heterogeneity

The learning rate ρ is expected to affect the class learning curve. The level of heterogeneity however affects the traditional mode of learning. Fig. 4(a) shows that the class learning curves belong to the sigmoidal plots except for characteristic slopes and shifts. As one would expect, high ρ_0 -values increase the initial number of learned students in both modes resulting to a relative shift in the onset of class overall learning. Such onset may be measured as the iteration number at which the fraction of learned students has reached 50% of the entire class. Opposed to TI, the shape of the learning curve for PI mode is more stable for different values of ρ_0 since only delay is introduced. TI's advantage for quick onset of learning however, is noticeable in all plots for the same ρ_0 -values.

The effect of varying ρ_0 is nonlinear. For both modes, a change of students' average learning rate from $\rho_0 = 0.1$ to 0.5 translates to a greater shift in the class learning curve, as opposed the same shift due to the change from $\rho_0 = 0.5$ to 0.9. Class-level improvements can therefore be affected by the initial average class performance.

The effect of class heterogeneity for both modes can be shown using a simple model with mixed student learning rates λ such that some are low ($\lambda - \delta\lambda$) and others are high ($\lambda + \delta\lambda$). Fig. 4(b) shows how this level of heterogeneity $\delta\lambda$ affects the resulting class learning curves. For the TI mode, the high level of heterogeneity at $\delta\lambda = 0.4$ (students have either $\lambda = 0.1$ or $\lambda = 0.9$ within the class) introduces a significant slow down midpoint of the initial rise of class learning curve. Such slow down is not significant for low heterogeneity. This two-stage learning phenomenon arises from the early transition of fast learners to catch up only during the later lecture iterations in Stage 2. The PI mode is robust to heterogeneity showing only slight uniform delay in class learning curve.

Spatial configurations of the class over iteration numbers provide insight on how learning spreads to the entire class. Fig. 4(c) shows the comparison of such configurations of the two modes over different iteration times but for smaller class size $N = 32$ for brevity. The first to learn with the TI mode are those who are fast learners (dark pixels with higher λ -values) and then spreads over the whole class later when those with lower λ -values eventually learns. A slow pickup in the learning curve is observable with PI mode. This phenomenon is due to the percolation-based spread of learned states within the class. The spread through all learners however appears to average out resulting to a sudden increase of class learning rate in the later iterations due to larger number of learning sites analogous to increased active sites in percolation phenomenon such as those in other CA-based models[25]. From these observations, we can

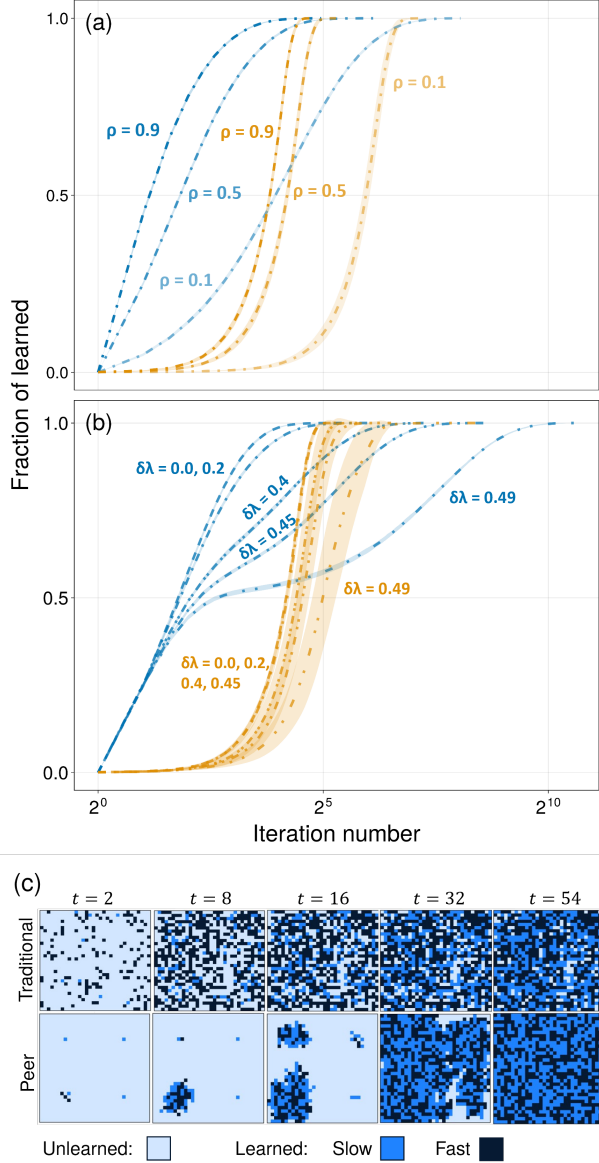


FIG. 4. Different monotonic increase of the total fraction of students in the learned state over iteration number. The blue lines represent the PI mode while the orange lines represent the TI mode. Unless indicated, the parameters are fixed such that $N = 64$, $\rho_0 = 0.5$, $\delta\lambda = 0.2$ (fairly homogenous) and an Inner Corner SA for PI mode. (a) Plots for varied ρ_0 -values, introducing varying degrees of shifts along the iteration (time) axis. (b) Plots for varied $\delta\lambda$ -values. (c) Representative development of classroom spatial configuration ($L = 32$, $\rho_0 = 0.3$) observed for different iteration numbers from $t = 2$ (each column doubles the previous t) for TI (first row) and PI (second row) modes. The learned and unlearned states are differentiated by the color intensity, with $\lambda = 0.9$ (darker) and 0.1 (lighter) shades.

also conclude that a mode using a combination of both TI and PI will benefit from these two phenomenon of inoculation of learning (via broadcast in TI step) and co-learning from peers (via percolation in the PI step). Current practices [1, 5, 7, 10] indeed also employ read-

ing materials in advance and the instructor gives a short lecture (TI mode) before the students engage in peer discussion (PI mode).

III.3. Combined effects

The combined effects of average transmission probability (learning rate) ρ_0 and of the level of heterogeneity $\delta\lambda$ is shown in Fig. 5 for two class sizes $N = 32^2$ (small) and 128^2 (large). The plots show the total time to learn $t_{\max}$ for the entire class. This time $t_{\max}$ is computed as the total number of iterations before the last cell (student) in the model has switched to learned state, or when the system does not change after about 20 rounds of random updates. For this plot, higher $t_{\max}$ is disadvantageous since it takes more iterations (hence time) before the entire class is expected to have learned.

Both Fig. 5(a) and Fig. 5(b) show that as the level of heterogeneity increases, the traditional approach (TI) require longer $t_{\max}$ than the peer instruction (PI) especially for small classes. The difference becomes smaller for higher ρ_0 or for higher average class aptitude (fast learners). For large classes, this difference can actually reverse, but only for cases of homogenous, fast learning, large classes. Classically, the utilization of class grouping (“sections” based on aptitude such as honors class) is thus best suited for TI. Instructional strategies would actually prefer PI only for cases of inhomogeneous class.

The same figure however also shows us that whenever we find TI advantageous, the difference in the $t_{\max}$ is not that large. In contrast, when PI is advantageous, the difference is very large. This suggests that PI can be a safe option —never performing much worse than TI, but can perform much better in many cases. The only disadvantage for PI is the huge delay in the rise of class learning curve. Overall, a mix of the two can be used to harness the advantages of both instruction modes.

IV. COMPARISON WITH AVAILABLE DATA

Compared to Roxas’ work [10, 11] which found the outer corner SA to perform the best, we found the inner corner SA to perform the best. This difference can be attributed to the simplifications made in our model. We were not able to consider the aptitude similarity effect that happens where students of similar aptitude learn better when grouped together regardless of their actual aptitude level [5]. The implementation of this phenomenon is better suited for a continuous-state model rather than a binary-state model. Additionally, we did not consider the orientation of the students in the classroom, resulting in an isotropic system. In addition to the findings regarding seating arrangements, our model also agrees with their findings that homogenous classes find better improvements compared to heterogeneous classes.

Our simulation design in its simplicity approximates

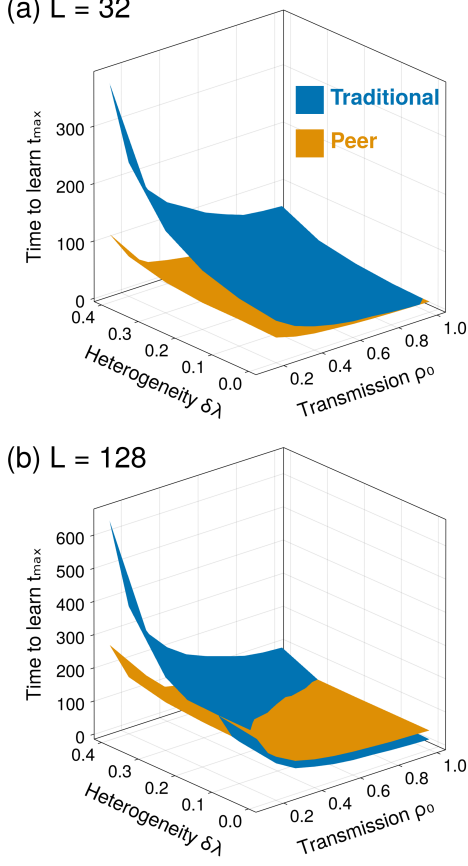


FIG. 5. Time to learn $t_{\max}$ as a function of positional learning factor ρ_0 and heterogeneity $\delta\lambda$ of PI (inner corner SA) and TI for two different class sizes $N = L^2$ ((a) $L = 32$ and (b) $L = 128$). The blue surface represent TI and the orange surface represents PI. Lower time to learn indicates better learning.

well other qualities of classroom learning by comparing our results with real-world data as follows. Class-level learning process is usually assessed via the Hake gain g and consider it related to the recurrence relation between successive evaluations S_1 and S_2 (cf. Equation 2). Since f can be used as an approximation for the probability S that the evaluation items are answered correctly by any member of the class, the Hake gain g can be used as a measure of how our CA model compares well with real-world data. Consequently, our model determines how g is dependent on S_1 by mapping the evaluation score to the fraction of learned students: $S \rightarrow f$, and that we have $S_2 = f_t$ for $S_1 = f_{t-1}$.

The Nitta Model can be considered as a recurrence equation of the item-based probability S of evaluation of the PI mode as shown to be in agreement with real-world experimental data [13]. The model being related to the Hake's gain model via $S \rightarrow f$ as above, can then be used to confirm the consistency of our simulation data with reality.

Nitta's model have been shown to agree with empirical data suggesting that a student who is likely to answer

an evaluation item correctly is more likely to answer any other item in the next evaluation. The gain function g can be approximated by the resulting function that fits the return map following Eq. 2. The return map that plots f_t against f_{t-1} can be shown to mimic Eq. 2 following the form of the recurrence equation $f_t = h(f_{t-1})$. The gain $g(S_1) \approx g(f)$ can then be approximated from real-world data (or simulation data) using

$$g(f) = \frac{h(f) - f}{1 - f} \quad (6)$$

with $h(f)$ as the fitting recurrence function to the return map.

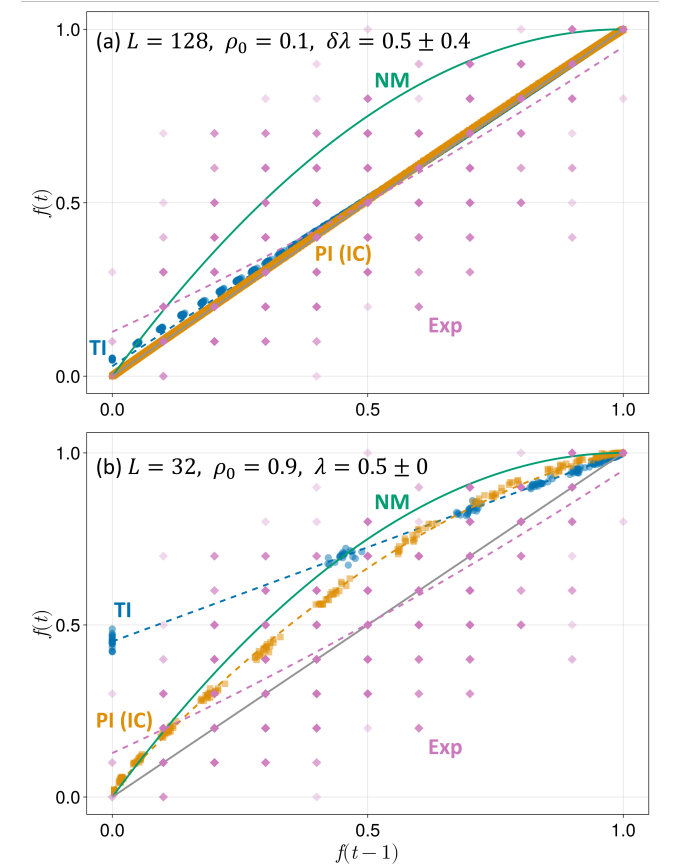


FIG. 6. Return map for selected parameter sets for both TI and PI. Blue circles represent the data points for TI and orange squares represent the data points for PI. Pink diamonds represent data points from Roxas' work [10] for comparison. The darker data points mean that more points overlap on those coordinates. The corresponding best-fit quadratic function obtained using the Levenberg-Marquardt algorithm for each set of data points are shown in dashed lines in their corresponding color and are discussed in the text. The green line represents Nitta's theoretical model (Equation 2) and the gray line represents $y = x$.

Fig. 6 shows the $h(f)$ that fits our simulation data for PI of smaller class sizes (Fig. 6(b)) approaching the expectations of the model of Nitta. We find that a

quadratic function fits [26] the return map for PI mode such that

$$h(f) = a_0 + [1 - (a_0 + a_2)]f + a_2 f^2 \text{ [PI mode]} \quad (7)$$

for some parameters a_i ($i = 1, 2$) with the condition that $h(1) = 1$. The TI mode for both cases always results to a linear trend [27] in $h(f)$ such that

$$h(f) = a_0 + (1 - a_0)f \text{ [TI mode]} \quad (8)$$

for a different parameter $a_0 = m$, again forcing the same limit for $f = 1$. We therefore see that

$$g(f) = a_0 - a_2 f \text{ [PI mode]} \quad (9)$$

$$g(f) = m \text{ [TI mode]} \quad (10)$$

for PI and TI modes, respectively (m replaced a_0 for the linear fit for distinction) [28]. When compared to Nitta's results [13], our simulation model shows $a_0 \approx 0$ making $g(f) \propto f$. Based on our model, PI mode results to a linear Hake gain consistent with their results but predicts a relatively constant gain for TI mode. The efficiency (Nitta) of the PI can also be related to the constant $|a_2|$.

In our simulation results (cf. Fig. 6), we find that the PI mode efficiency is lost to high class size giving $|a_2| = 0.60$ for $N = 32^2$ to 0.03 for $N = 128^2$. TI mode gain also declined from $m = 0.45$ to 0.029 with the same increase in class size. These results are consistent with the overall assessment based on other dynamical parameters presented above.

Our results are consistent with the empirical data reported by Roxas [10]. Roxas' dataset, consists of pre- and post-test scores from fourteen PI classrooms and spans a broad range of values. Fig. 6 plots the pre-test scores plotted as f_{t-1} and post-test scores plotted as f_t . The variety in the data can be attributed to many factors that affect classroom learning dynamics such as instructional strategies which may affect the dominance of the effects of TI over PI. The timing of pre- and post-tests and the level of precision of the assessment method can also influence the observed outcomes. Roxas' dataset includes instances of score decreases from pre- to post-test—a realistic effect not captured by our model. We obtain coefficients of the quadratic terms— $a_2 = 0.14$ for Roxas [29], $a_2 = -0.033$ for our unfavorable case, $a_2 = -0.605$

for our favorable case, and $a_2 = -1$ for Nitta. Taken together, these results indicate that while our PI model does not reproduce all empirical features, it provides a close approximation to the observed classroom dynamics shown in Nitta's model and Roxas' data.

V. SUMMARY AND CONCLUSION

We propose a probabilistic cellular automata model as a new way of investigating learning dynamics in the classroom. Through this model, we investigated how different factors affecting classroom learning dynamics for both traditional instruction (TI) and peer instruction (PI).

We found that TI performs better in larger classes and when the students have low learning rate heterogeneity. On the other hand, PI performs better in smaller classes and when the students have high learning rate heterogeneity. Larger class size (N), higher level of heterogeneity ($\delta\lambda$), and higher learning transmission efficiency (ρ_0) are not advantageous to both TI and PI modes of classroom learning. TI mode though is not as affected by class size compared to PI while heterogeneity has lesser impact with PI mode. Other simulation observations are discussed consistent with the real-world experience on these modes of classroom instruction.

Consistent with known empirical real-world data results, our simulation data is able to capture theoretical predictions of the possible forms of Hake gains for both TI and PI modes of the model. We presented an empirical determination of the effective Hake gain of the mode of instruction based on a return map data from class level pre- and post-test results. This approach will provide future studies better way of determining class-level Hake gains instead of solely based on individual averages.

While our model is simplistic, we have shown that it be a good approximation of the real world and higher-order effects be easily incorporated (e.g. anisotropy of learning in PI, more levels of learner types) The current model only considers the peer discussion part of a whole PI session. A mixed model that considers the whole session with a mix of TI and TI would be a better representation of the real world. Additionally, transitioning to a continuous-state model would allow incorporating more real-world phenomena like aptitude similarity and orientation effects during co-learning.

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- [26] $f_t = -0.033f_{t-1}^2 + 1.033f_{t-1}$ in Fig. 6(a). $f_t = -0.605f_{t-1}^2 + 1.595f_{t-1} + 0.019$ in Fig. 6(b).
- [27] $f_t = 0.97f_{t-1} + 0.029$ for Fig. 6(a) and $f_t = 0.548f_{t-1} + 0.452$ for Fig. 6(b).
- [28] For any polynomial function of degree N fitting the return map, the Hake gain takes the general form $g(f) = a_0 - \sum_{n=2}^{N-2} a_n f \cdot \sum_{m=0}^{n-2} f^m$.
- [29] $f_t = 0.14f_{t-1}^2 + 0.68f_{t-1} + 0.13$.